Status	Finished
Started	Wednesday, 7 October 2026, 9:56 PM
Completed	Wednesday, 7 October 2026, 10:03 PM
Duration	7 mins 24 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Correct

Mark 1.00 out of 1.00

What is the key characteristic of binary search?

a. It works on unsorted lists
b. It compares elements sequentially
c. It always starts from the beginning of the list
d. It can be applied only if the list is sorted ✓

Question **2**

Correct

Mark 1.00 out of 1.00

In binary search, what happens if the middle element does not match the target element?

a. The search continues from the beginning
b. The list is sorted
c. The search stops
d. The search continues in the left or right sublist ✓

Question **3**

Correct

Mark 1.00 out of 1.00

Which of the following is not a limitation of binary search algorithm?

a. Binary search algorithm is not efficient when the data elements more than 1500 ✓
b. There must be a mechanism to access middle element directly
c. Requirement of sorted array is expensive when a lot of insertion and deletions are needed
d. Must use a sorted array

Question **4**

Correct

Mark 1.00 out of 1.00

Which of the following is not the required condition for a binary search algorithm?

a. The list must be sorted

b. Number values should only be present

Number values should only be present

c. There must be a mechanism to delete and/or insert elements in the list ✓

d. There should be direct access to the middle element in any sublist

Question **5**

Correct

Mark 1.00 out of 1.00

During a binary search, what happens if the target element matches the middle element?

a. The list is sorted

b. The search ends successfully ✓

c. The search continues in the right sublist

d. The search continues in the left sublist

Question **6**

Correct

Mark 1.00 out of 1.00

If a list contains 1000 elements, how many comparisons would a binary search typically make in the worst case?

a. 1000

b. 10 ✓

c. 500

d. 100

Question **7**

Correct

Mark 1.00 out of 1.00

What is searching in the context of computer science?

a. Inserting elements into a list

b. Sorting elements in a list

c. Determining whether an element is present in a list ✓

d. Deleting elements from a list

Question **8**

Correct

Mark 1.00 out of 1.00

What is the first step in binary search?

a. Compare the target element with the first element in the list
b. Compare the target element with the middle element in the list ✓
c. Sort the list
d. Divide the list into two equal parts

Question **9**

Correct

Mark 1.00 out of 1.00

In a linear search, how many comparisons are made in the worst-case scenario to find an element in a list of size n?

a. log n
b. n/2
c. 1
d. n ✓

Question **10**

Correct

Mark 1.00 out of 1.00

What is the time complexity of linear search in the worst case?

a. O(1)
b. O(n log n)
c. O(log n)
d. O👎 ✓

Question **11**

Correct

Mark 1.00 out of 1.00

Finding the location of a given item in a collection of items is called

a. Searching ✓
b. Finding
c. Discovering
d. Mining

Question **12**

Correct

Mark 1.00 out of 1.00

If the target element is greater than the middle element in binary search, where does the search continue?

a. In the left sublist
b. At the beginning of the list
c. In the middle of the list
d. In the right sublist ✓

Question **13**

Correct

Mark 1.00 out of 1.00

In binary search, how is the middle element determined?

a. By dividing the list length by two ✓
b. By using a hash function
c. By comparing each element sequentially
d. By starting from the first element

Question **14**

Correct

Mark 1.00 out of 1.00

Which of the following is a limitation of binary search?

a. It requires the list to be sorted ✓
b. It can only be applied to large lists
c. It does not work with negative numbers
d. It is slower than linear search for small lists

Question **15**

Correct

Mark 1.00 out of 1.00

Given an array arr = {45,77,89,90,94,99,100} and key = 99; what are the mid values(corresponding array elements) in the first and second levels of recursion?

a. 90 and 99 ✓
b. 90 and 94
c. 89 and 94
d. 89 and 99